LEGENDS

VALVE SYMBOLS

- VRE = CHECK VALVE
- VES = BALL VALVE
- GT = GATE VALVE
- VES = AKTUATOR BALL VALVE

- LS = LEVEL SWITCH
- FM = FLOW METER
- AE = SENSOR / ANALYZER
- PHC = PVC
- GIP = GALVANIZED IRON PIPE

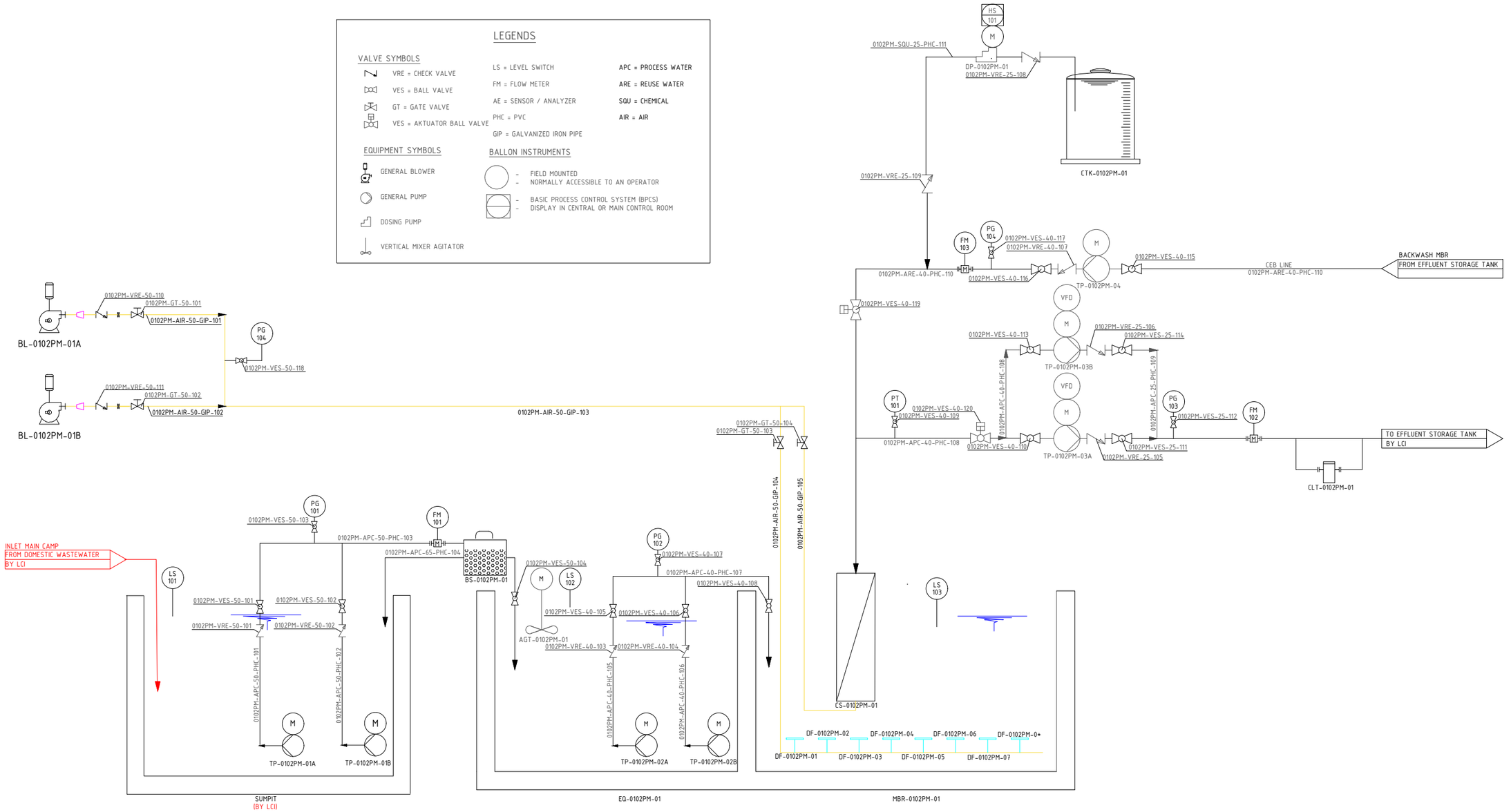
- APC = PROCESS WATER
- ARE = REUSE WATER
- SQU = CHEMICAL
- AIR = AIR

EQUIPMENT SYMBOLS

- GENERAL BLOWER
- GENERAL PUMP
- DOSING PUMP
- VERTICAL MIXER AGITATOR

BALLON INSTRUMENTS

- FIELD MOUNTED
- NORMALLY ACCESSIBLE TO AN OPERATOR
- BASIC PROCESS CONTROL SYSTEM (BPCS)
- DISPLAY IN CENTRAL OR MAIN CONTROL ROOM



TAG.NO	TP-0102PM-01A/B
EQUIPMENT	SUMPIT TRANSFER PUMP
TYPE.	SUBMERSIBLE PUMP
CAPACITY	2.8 CMH HEAD 4.8 m
POWER	0.37 kW/1 PHASE/230 V
QTY.	2 UNITS (1W+1S)

TAG.NO	TP-0102PM-02A/B
EQUIPMENT	EQ TRANSFER PUMP
TYPE.	SUBMERSIBLE PUMP
CAPACITY	2.25 CMH HEAD 4.47 m
POWER	0.37 kW/1 PHASE/230 V
QTY.	2 UNITS (1W+1S)

TAG.NO	TP-0102PM-03A/B
EQUIPMENT	MBR PERMEATE PUMP
TYPE.	VERTICAL MULTISTAGE PUMP
CAPACITY	2.2 CMH HEAD 10 - 15 m
POWER	0.37 kW/3 PHASE/380 V
QTY.	2 UNIT

TAG.NO	TP-0102PM-04
EQUIPMENT	MBR BACKWASH PUMP
TYPE.	VERTICAL MULTISTAGE PUMP
CAPACITY	3.89 CMH HEAD 15 - 20 m
POWER	0.75 kW/3 PHASE/380 V
QTY.	1 UNIT

TAG.NO	BL-0102PM-01A/B
EQUIPMENT	MBR BLOWER
TYPE.	ROOTS/SIDE CHANNEL BLOWER
CAPACITY	2,2 Nm3/min
POWER	2.2 kW/3 PHASE/380 V
QTY.	2 UNITS (1W+1S)

TAG.NO	DP-0102PM-01
EQUIPMENT	CEB DOSING PUMP
CAPACITY	80 LPH 10 BAR
POWER	0.37 kW
QTY.	1 UNIT

TAG.NO	EQ-0102PM-01
EQUIPMENT	EQUALIZING TANK
CAPACITY	5,4 M3
MATERIAL	MILD STEEL
QTY.	1 UNIT

TAG.NO	MBR-0102PM-01
EQUIPMENT	MEMBRANE BIO REACTOR TANK
CAPACITY	14,2 M3
MATERIAL	MILD STEEL
QTY.	1 UNIT

TAG.NO	CS-0102PM-01
EQUIPMENT	CASSETE MEMBRANE
MATERIAL	PVDF

TAG.NO	CTK-0102PM-01
EQUIPMENT	CIP CHEMICAL TANK
CAPACITY	25 L
QTY	1 UNIT

TAG.NO	CLT-0102PM-01
EQUIPMENT	CHLORINE TABLET
TYPE	1 UNIT

TAG.NO	BS-0102PM-01
EQUIPMENT	BASKET SCREEN
TYPE	1 UNIT

TAG.NO	DF-0+
EQUIPMENT	DIFFUSER AERATION
TYPE	18 UNIT

TAG.NO	AGT-0102PM-01
EQUIPMENT	AGITATOR MIXER
POWER	0,55 kW
QTY	1 UNIT

NOTES

03 JUNI 2025_REV.0_FIRST ISSUED
05 DESEMBER 2025_REV.1_ISSUED FOR REVISION
22 JANUARI 2026_REV.2_ISSUED FOR REVISION

REFERENCE DOCUMENTS

REV.	I.T.	DESCRIPTION	BY	DWG.	CKD.	APPR.	AUTH.	DATE
A	A	ISSUED FOR REVIEW	TA	TA	AZ	DW	OR	05/12/2025

REVISIONS

I.T. ISSUE TYPE: (A) PRELIMINARY (B) FOR APPROVAL (C) FOR INFORMATION (D) FOR QUOTATION (E) FOR CONSTRUCTION (F) AS PURCHASED (G) AS BUILT (H) CANCELLED

VALE | LEIGHTON

CLASSIFICATION: CONFIDENTIAL

PROJECT: IGP – INTEGRATED MINES POMALAA (IMP)
PROJECT No. OR SE:
CONTRACT No.: –

DETAILED DESIGN PROJECT
OPERATION CAMP AREA
WASTE WATER TREATMENT PLANT
PIPING & INSTRUMENTATION DIAGRAM (PID)

SCALE	CONTRACTOR'S No.	VALE No.	REVISION
–	–	DF-FG-0102PM-P-94206-22	A